Porting Android Automotive on Xen

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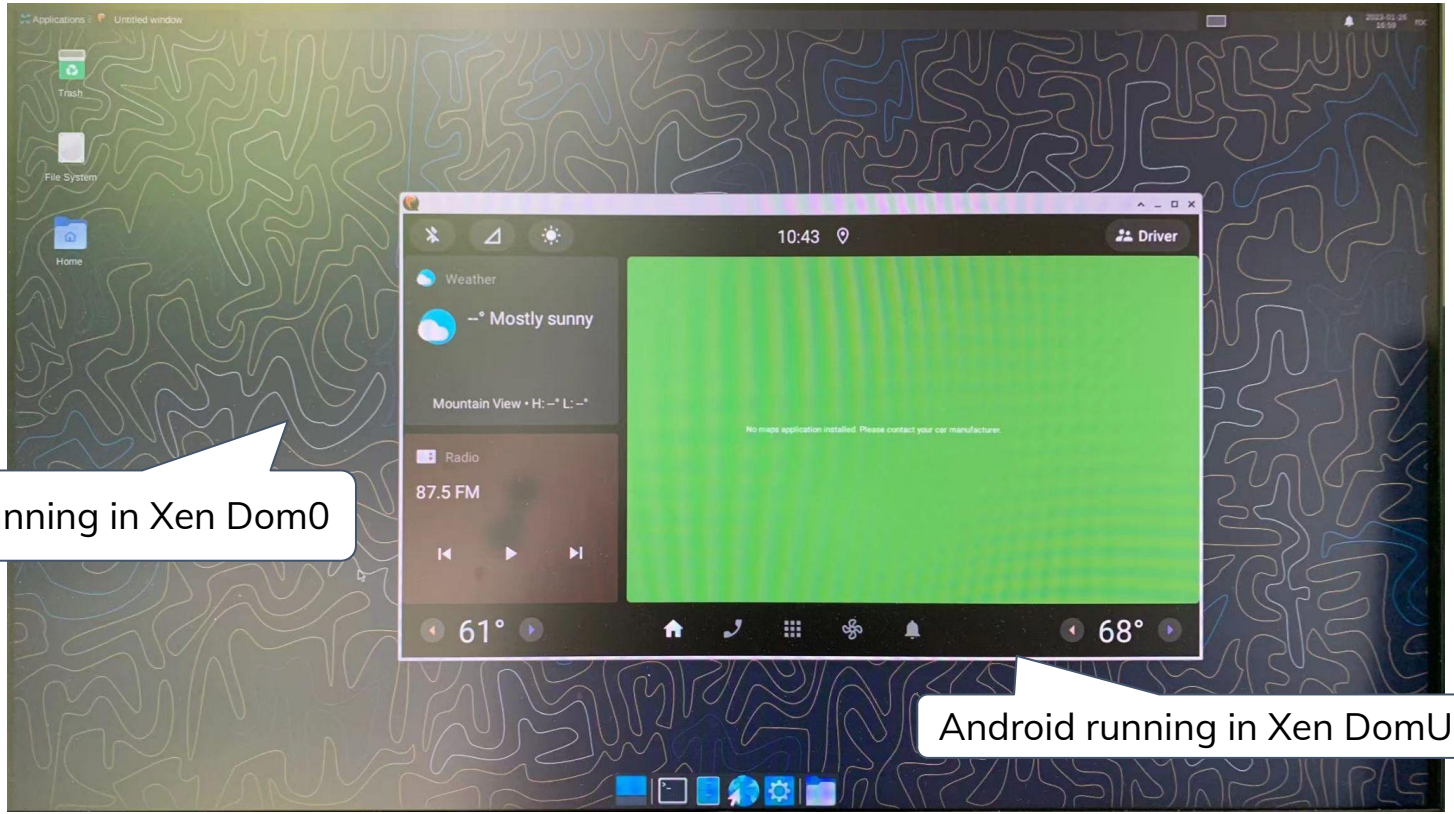
Introduction

- As a part of automotive project, the goal is to identify the gaps associated with running Android Automotive OS in a virtual machine using the Xen hypervisor.
- The objective was to leverage the Android Cuttlefish VM as a Xen hypervisor guest with using virtio based virtual device emulations. That involved:
  - Host is Yocto based distro: TRS or SoC vendor's distro (includes Xen, Linux kernel, and QEMU, etc)
  - Guest is Android Open Source Project (AOSP), built on master branch with `aosp_cf_arm64_auto-userdebug`* target
  - The proof of concept is in part based on work done by EPAM
- Tested on 3 hardware platforms:
  - ADLink AVA Platform (Ampere Altra SoC, 32 Arm Neoverse N1 CPUs)
  - Telechips Dolphin5 automotive platform (Arm Cortex-A76 x 4 + Cortex A55 x 2)
  - Rockpi 4b (Arm Cortex A72 x 2 + Cortex A55 x 4) (Jun Nie)

* It was renamed as `aosp_cf_arm64_auto-trunk_staging-userdebug` in Oct, 2023

Screenshot ;-)



Linux running in Xen Dom0

Android running in Xen DomU

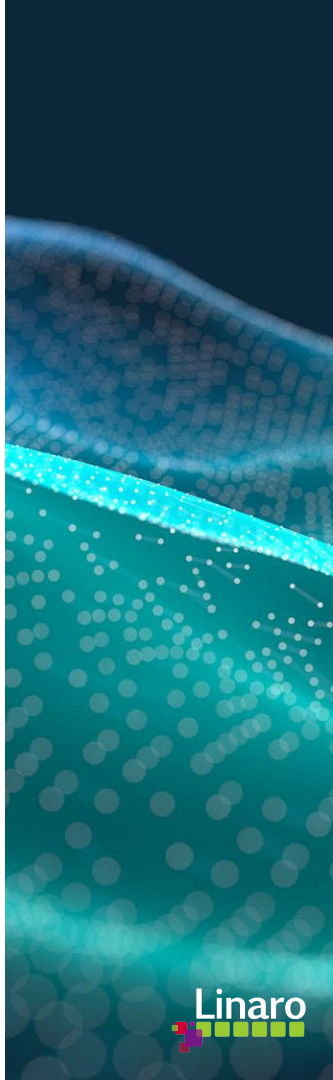
Linaro

Current status

- Support Xen virtual machine in Android U-boot [patches - merged]
- Support Xen fundamentals and frontend drivers in Android common kernel [~~patch~~ - no plan for upstreaming]
- Support Xen virtual machine in Android Cuttlefish device folder (`$android/device/google/cuttlefish`)
  Set display finder mode as “drm” [patch - merged]
  Support booting on Xen virtual block device [patch]
  Fix console hvc0 regression [patch]
- Maintain Xen and QEMU repositories for support virtio devices on Xen / Arm64
- Known issue: Cuttlefish’s secure_env is absent on Xen, file system encryption, oemlock and other security services have been disabled.

- Demo: Booting Android automotive OS on Xen virtual machine

Next steps

- Upstream Android Cuttlefish patches for better support Xen virtual machine.
- Support virtual devices with vhost based backends
  - Develop vhost user based backends in a hypervisor-agnostic way: vhost-device
    RNG, I2C, GPIO, SCSI, sound (WIP), input (WIP), etc.
  - Vulkan support for virtio-gpu
- Any suggestions are welcome!

Thank you!